

Topological Data Analysis of High Speed Liquid Impact

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Abstract

The use of three-dimensional point clouds are a recent addition to the toolkit of techniques for analysing high speed liquid impacts. With the improvement in experimental imaging capabilities and associated modelling and numerical work within high performance computing architectures, developing techniques to utilise this potential is of great importance. This report, therefore, seeks to analyse datasets of this type, specifically the canonical, physical scenarios of a droplet impacting a deep pool and two droplets of the same radius colliding at high speeds, in order to compare to analytically derived theories for properties of secondary ejected droplets and the topological features of the jet formed from the impact. Notably, it is found that the radii of the ejected droplets are within bounds found in previous work by other authors, the droplets have a positive correlation between velocity and volume within each time frame, and the growth of the roots of the jet away from the y -axis as well as the distance between the roots in the case of a droplet impacting a deep pool grow with a power of t . Additionally, developing theory that the thickness of the jet evolves with a power of r , the distance from the y -axis, is shown to have strong evidence in the case of two impacting droplets, and the distance between the roots in this scenario appears to grow with a power of t , a previously undiscovered behaviour.

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I would also like to thank my cat, Smidge, for meowing at me whilst I wrote this.

I hereby certify that this dissertation has been written by me, and that it is the record of work carried out by me, or principally by myself in collaboration with others as acknowledged, and that it has not been submitted in any previous application for a higher degree. The length of the report does not exceed 12,000 words.

I hereby declare that Generative AI tool Claude Sonnet 5 [6] was used in a minor amount of the code in order to perform menial tasks. The code created is present in the accompanying materials and any functions which were AI assisted have been clearly labelled as such in their docstrings. This code was fully understood before using and was subsequently edited by me in order to better curate the code to my needs.

1 Introduction

Numerical simulations have been increasingly used over all areas of mathematical physics during the past decades and they have only become more prominent as computational power has increased. Within computational fluid dynamics specifically, a great area of interest for these capabilities has been in impact theory of fluids, specifically in the analytic formulation of Wagner theory [27] which focuses on liquid impacts where the Reynolds number is high. Specifically, large datasets where interfacial shapes can be represented via three-dimensional point clouds are increasingly being used for such simulations and their subsequent analysis [28].

This report aims to create a methodology to harness these large datasets to garner mechanistic insight into the dynamics of violent droplet impact. Doing so will help improve intuition about the splashing process and potentially inform both experimental and analytic campaigns. In order to test this methodology, the outcomes are compared to analytic predictions from Wagner theory as well as results from previous simulations and image analysis of droplet splashing in the canonical cases of a droplet landing on a pool and two droplets of the same volume directly colliding, both at high speeds. In this report, statistics of the secondary ejected droplets from the splash are collected and analysed, including the droplet count, their velocities and their volume distributions; a method is devised to analyse a three-dimensional point cloud of the interface of the splash by projecting individual slices into two-dimensions, the roots of the jet (see [4]) are found and their movement over time analysed, as well as the thickness of the jet through time and space.

This study begins with the general background of the research area and more recent advancements in the field before describing the methodology created to perform the subsequent analysis. First describing how the secondary droplets were analysed before moving onto the interface analysis, highlighting the techniques to identify the roots of the jet, the points along the jet itself and different methods to find the thickness of the jet. After that, the results from the analysis are presented along with a discussion of their significance and a critique of the methodology are laid out in the next section. Within the methodology critique, suggestions are laid out for future work before the report is concluded. A GitHub repository of accompanying materials is available [14] in which the code created to analyse these materials as well as some animations are available to view.

2 Background

In the 1930s, work by Wagner [27] and von Kármán [26] was carried out to produce mathematical frameworks to model liquid impact problems on large scales. Understanding the forces in these situations was important for the design of aeroplanes which could land in the sea as well as for protecting the hulls of large ships as they slammed against waves; these situations are what the original pursuit of such knowledge was prompted by. This began the study of liquid impacts and the creation of techniques to analytically assess such phenomena, which became useful in the design of oil rigs in the sea [1]. In the later 20th century, this work was adapted to focus on smaller impact scales by the likes of Korobkin & Pukhnachov [13], instead focusing on singular small droplets and the more intricate geometries which arise from their splashes. The flexibility of this theory has been shown as it has been used in a wide range studies from inkjet printers to asteroid impacts [10]. An especially pressing area is in aircraft de-icing: whilst plane crashes are thankfully rare, icing issues are one of the most common causes of flight anomalies when they do occur. The icing is caused when a plane flies through a cloud and water droplets collect on the wings, as the plane ascends in altitude and the temperature falls, water still stuck to the wing can turn to ice, therefore understanding how much liquid remains on a solid surface after a high-speed impact is paramount. Image analysis has been used to complement the analytic theory, Burzynski et al. [2] did this in the case of a liquid impacting on a dry surface, studying the volume and velocity distributions of the ejected droplets over time which was useful in verifying many of the analytic predictions from the field.

In more recent years, there has been an introduction of computational fluid mechanics into this area of study. A computational approach to the splashing on a dry surface was carried out by Cimpeanu & Papageorgiou [5] which found similar results to both analytic and image findings, justifying that the computational software was powerful enough to be used. Also studied have been impacts of droplets onto liquids; the effects of a droplet on a thin film of liquid were derived analytically by Howison et al. [10] and the effects studied numerically by Josserand et al. [12]. The same techniques have been applied to the analysis of droplet

impacts on a deep pool, Murphy et al. [18] analysed this using a high speed camera and the same set-up was then used by Wang et al. [28] numerically. This allowed otherwise unattainable perspectives of the splash mechanisms where imaging is difficult and analytic theory breaks down. Intricate patterns were found in a similar set-up by Thoraval et al. [25] who found von Kármán vortex streets within the splash showing that there is much to be uncovered in this world.

Whilst image analysis of splashes have proved to be of great use, it can be difficult to extract the finest details of the splashing phenomena such as the smallest of the ejected droplets and intricacies of the geometry involved at early impact times. In the paper by Cimpeanu & Moore [4], predictions of details such as the evolution of the roots of the jet and the thickness of the jet were made analytically before being tested numerically in two dimensions. Using image analysis to test such theories would be incredibly difficult since the jet is a two-dimensional slice of the three-dimensional splash sheet, so extracting its intricate details would be incredibly challenging especially at its base since the rest of the liquid gets in the way of the points of interest. Understanding this is especially important for oil spills in the sea. If there is a leak of oil from a rig out at sea, the oil floats at the surface of the water, if it then rains then the splashing will spread the oil further leading to a wider range of pollution. Understanding the specifics of splash geometry for each individual drop to the finest details will make simulations far more reliable, since improving the simulation of one raindrop will make the total simulation with thousands of raindrops considerably more insightful.

The use of three-dimensional point clouds to analyse such features is also important to develop. In contrast to an image where pixels are arranged on a regular grid, point clouds do not inherently have any such structure [17] (page 2). For example, if you shuffled the order of the pixels for an image of a cat and attempted to display it then you would get something entirely incoherent since each pixel goes to a specific coordinate. However, if a model of a cat came in the form of a three-dimensional point cloud, a cat would still be displayed no matter what order the points were presented in since the information itself is a set of coordinate points. This makes finding particular regions of interest more difficult since the data for this region could be scattered entirely randomly across the full dataset, unlike an image where it would be known which pixels correspond to which area of the image a priori. This is a central problem for AI models which attempt to classify objects from point cloud information and many architectures have been developed to tackle this such as PointNet [3], DGCNN [29] and PointCNN [16], though these are not the focus of this study.

The motivation of this study, therefore, is to analyse numerical data of high-speed liquid impact simulations using three-dimensional point clouds. The theory in this area is well grounded in two-dimensions, with three-dimensional analysis techniques being developed, the work in this report is another step towards developing the three-dimensional analysis capabilities further. Previous analysis has been illuminating but has been done only analytically, using images, numerically but only in two dimensions, or has used three-dimensional point cloud simulations but focused on a far larger time span. The analysis by Wang et al. [28], for example, simulated a splash for over 40 ms after impact. This study focuses solely on the very early impact stage, lasting no more than 20 μ s after impact.

3 Methodology

The data studied throughout this report was produced using the open source BASILISK software [19] [22] [20] [21] to create a total of five simulations for two different scenarios: a droplet splashing on a pool of water and two droplets of the same size and volume colliding into each other. The scales of these simulations are equivalent to water droplets of diameter 1 mm impacting at 10 ms^{-1} , for specifics on the initial conditions of these simulations see Appendix A (6.1). However, all quantities were made relative before being passed into the software by using dimensionless parameters which have no units, this helps to make comparisons across scales more straightforward. Therefore, throughout this report only the numerical values will be reported on since the values themselves are of less importance than their general behaviour, other than the value of time for which one unit of time is equivalent to 5×10^{-5} s. The five simulations consisted of three simulations of a droplet on a pool, one with a lower resolution and two at a higher resolution, and two of the droplets colliding with one at the lower resolution and the other higher. The lower resolutions are at level $L = 9$ whilst the higher ones are at level $L = 10$. This means that there are at most 2^L grid cells in each dimension in which the fluid particles' positions can lie; however, in order to prevent 2^{3L} grid cells being used in total, which would require immense computational power, the program uses adaptive time stepping meaning that the size of grid cells in a local area depends on the position and velocity of the fluid particles. We can see

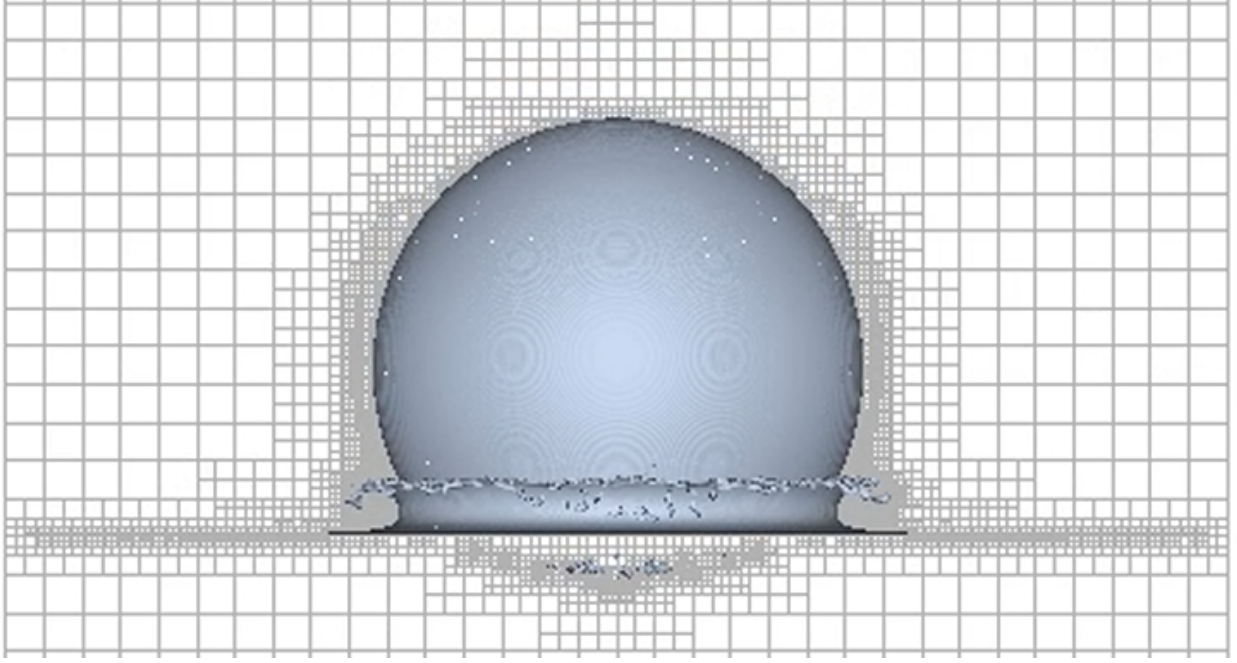


Figure 1: The grid cell sizes changing depending on the water droplet features.

how the size of the grid cells change in Figure 1. The lower resolution simulations took approximately one day on four CPUs to run whilst the higher resolution simulations took around three days on eight CPUs.

The data created was three-dimensional, with gravity and the droplet's initial velocity acting in the y -direction, a strategy used to aid analysis was to take two-dimensional radial slices from the data which kept the y -values the same but combined the x - and z - coordinates to represent distance from the y -axis. This is referred to as x^* in this report, with $x^* = \sqrt{x^2 + z^2}$.

3.1 Ejected Droplet Analysis

The first part of this project focused on analysing the smaller ejected droplets which appear after the impact of liquids. The data in this section was taken from the three-dimensional point cloud simulations of the splashes with the data converted to identify contiguous bodies of liquid complete to give the data in the form of the timestamp of the frame, the Cartesian coordinates of the centres of mass for each of these bodies, their volumes, and the velocities of the centres of mass also in Cartesian coordinates. Table 1 gives some example data, the full size of these datasets ranged from around 1300 rows for the lower resolution simulations to 4250 for those using a higher resolution, specifics on how the data should be laid out can be found in Appendix B (6.2).

Frame	Timestamp	Droplet ID	Volume	x	y	z	v_x	v_y	v_z
0	0	0	62.5	2.5	1.25	0	0	0	-3.196e-11
0	0	1	2.094	0.3750	3.7	-8.263e-17	1e-4	-1	-3.126e-8
11	0.01	0	62.499	2.5	1.25	-3.196e-11	0	0	-3.196e-11
11	0.01	1	2.094	0.375	3.690	-3.126e-10	1e-4	-0.999	-3.126e-8

Table 1: First 4 rows of actual data used.

For this report, a number of visual tools were created to extract relevant statistics from these datasets. These include graphs for the number of droplets over time, the average speed of the droplets over time, a three-dimensional graph where each slice is the log of the droplet volumes against their frequency at each

time step, as well as a three-dimensional histogram of the droplet volumes over time, and a scatter-plot of the log of the droplet volumes against their speeds.

In order to get the time dependent statistics, the data from each time step must be isolated. This is performed by taking only the rows of the DataFrame with a `time` value within a tolerance of 1×10^{-4} . The studied simulations had time steps of 1×10^{-2} so this tolerance would properly isolate the frame whilst also accounting for any floating point errors picked up when transferring the data from the file to the DataFrame. For each time step, statistics such as the number of droplets can be found by simply taking the length of the subset, and the average speed is found by calculating $\sqrt{v_x^2 + v_y^2 + v_z^2}$ for each droplet and dividing by the total number of droplets. Since the simulations are three-dimensional and have droplets ejected in all directions the speed must be used rather than the velocity, else droplets fired in different directions could cancel each other out. These values are all appended to a list from which they can be graphed against the time steps which gives the graph of the evolution of these statistics over time.

A three-dimensional histogram is a useful tool to study the evolution of the droplet volume distributions over time. Since the volume of the ejected droplets scale over multiple orders of magnitude, a log scale is needed to study these distributions properly. To create this figure, the droplets are again separated into individual time steps, a histogram is then created with an equal number of bins for each one (it was found that `n=50` worked best for the datasets studied) and these are put together into a three-dimensional bar graph with each bar coloured based on its density, a density of 0 being set to blue and the most to orange with a gradient for values in between. The data had to be reduced to only consider droplets smaller than the original bodies and to ensure that some time has passed since collision such that enough droplets have been ejected to not skew the histogram, otherwise the colour gradient is dominated at the very instance of collision when only a handful of droplets have been ejected and the colour gradients become indistinguishable for the later times when more droplets have been ejected. Two angles are then given: one view with an elevation of 30° and an azimuth of 60° , the other being a bird's eye view to better see the full distributions. The same is also available for a frequency histogram, the code is almost precisely the same but `density = False` is selected when creating the histogram. This helps show the growth in the number of droplets over time as well as their distributions which will be compared to analytic predictions and previous experiments such as those by Burzynski et al. [2] in the next section. Notably, the evolution of the average droplet velocity over time as well as the distributions of the ejected droplet volumes.

Due to the size of the true datasets, tests were carried out on a self made miniature model which could produce synthetic data of the same structure but without the computational burden of a full fluid simulation. Details can be found in Appendix C 6.3.

3.2 Interface Analyser

The second and larger part of this project focused on building frameworks to identify regions of interest from a three-dimensional point cloud simulation of a splash. Specifically identifying the jet itself beginning at its roots, the movement of the root positions over time, the evolution of the line running through the middle of the jet, and the change in thickness of the jet through space and time. The data is filtered to show only liquid points at the boundary between the contiguous fluid and the air. The two-dimensional slice of one of these datasets with $z = 0$ is displayed in figure 2. All methods were originally devised on a dataset of a droplet splashing onto a pool and then adapted for the situation of two droplets colliding later, thus the former is the image that should be kept in mind when the evolution of the methodology is spoken about, the adaptations made for the two droplets are explained afterwards.

3.2.1 Projection to 2 Dimensions

The features of interest have definitions which relate to neighbouring points such as curvature and gradients which are difficult to make robust in three dimensions since the nearest neighbour may lie in the z -direction when it was desired to traverse in the y -direction. Moreover, the full datasets are incredibly large, containing over two million points for each frame at the high resolution. This is too high for the graphing software Matplotlib [11] to plot on three-dimensional axes so reduction into smaller parts was necessary. The first task was, therefore, to create a method to reduce the dataset into two-dimensional slices which could be more easily analysed. Since the droplets are spherical and their splash regions are circular or elliptical in nature,

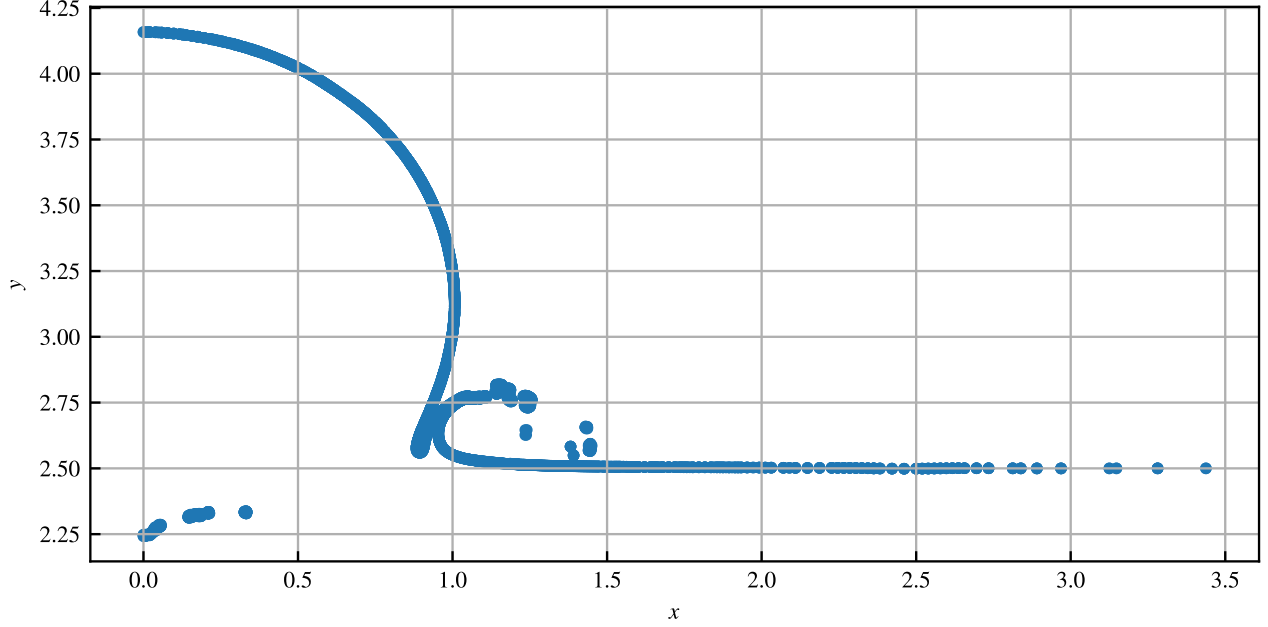


Figure 2: Two-dimensional slice of three-dimensional interface.

it made intuitive sense to create these slices radially rather than along Cartesian coordinate lines. However, the simulations are created using a grid structure so any given line may not pass through enough points to produce a usable two-dimensional interface to analyse. Thus, the solution made was to form a radial 'slice' centred at the origin and to take all points within an angle range. These points were then projected onto a single line which could then be analysed more easily. The projection is performed as follows: a slice from, say, 44° to 46° above the origin is taken and these points will be projected onto the line which bisects this slice, that being 45° above the origin. This line can be represented in polar coordinates as the line

$$(r \in [0, \infty), \theta = 45^\circ).$$

Now all points in this slice are represented as

$$(r_i, \theta + \phi_i),$$

to project this onto the line we simply convert this coordinate to be (r_i, θ) as follows:

$$r_i = \sqrt{x^2 + z^2}, \quad x_i = r_i \cos(\theta), \quad z_i = r_i \sin(\theta),$$

with the y -coordinate remaining the same. Figure 3 is a visual aid for this process.

With a small enough slice, the points will be radially symmetric which may mean that multiple points get projected on top of each other. This again makes analysis tricky if not done by hand since the methods made use nearest neighbour searches to traverse along the outline of the splash. Therefore, duplicated points are defined to be any points which have coordinates within a Euclidean distance of 1×10^{-3} of each other and any points within that range of another point are removed leaving only the point with the smallest index from the DataFrame. A distance of 1×10^{-3} was chosen since this was large enough to remove all duplicates and small enough that it would keep a large portion of points when applied to the original DataFrame. Any points which it did remove from the original data were commonly troublesome during nearest neighbour searches anyway and could cause the code to terminate prematurely.

3.2.2 The Roots of the Jet

Of particular interest are the so-called roots of the jet. These are defined to be the locations at the base of the ejected liquid where the curvature is locally maximised, see [4]. The locations of the roots are shown for

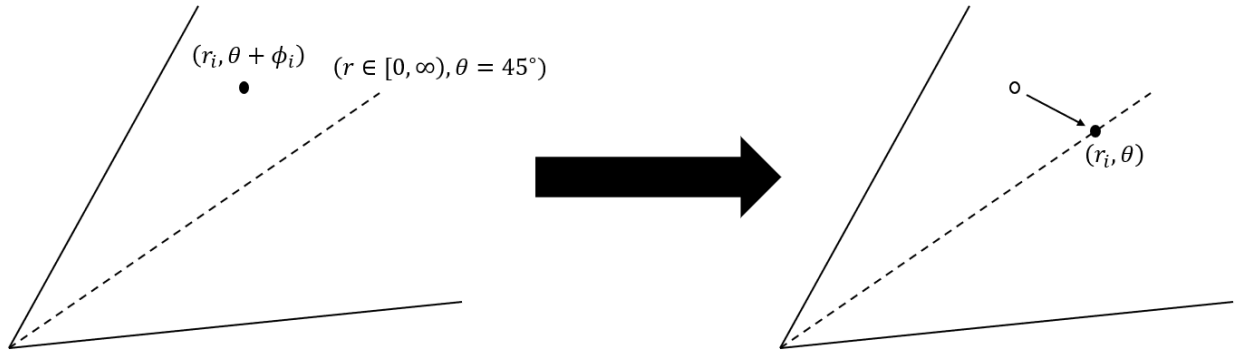


Figure 3: Projection of a point to the line.

each studied case in figure 4. Throughout the report, these features are referred to as the left and right roots for the droplet on a pool case, referring to the root closer to the y -axis and the root further from the y -axis in the image on the left respectively, and the top and bottom roots for the two colliding droplets, referring to the roots with a higher and lower y value in the image on the right respectively. Since the coordinate x^* is used in this report, all images will be oriented in the same direction so there will not be any confusion referring to left and right roots with a mirror image of figure 2, for example. In the literature they are sometimes referred to as the $+$ and $-$ roots in order to be more precise but the left and right/top and bottom terminology gives a more direct mental image of what is being referenced so have been chosen for this report.

Originally, the approach to finding these roots was to perform a nearest neighbour search along the interfaces by building a k -d tree [23] and traversing it, then outputting the points with the highest curvature. The curvature of a point was found by taking its coordinates and the coordinates of the points either side of it and then calculating their Menger curvature [15]. However, this ran into issues since there was usually more curvature at the tip of the jet than at the roots, so stopping criteria were introduced for the search: when the points started going upwards for the root on the left and when the points started going to the right for the root on the right. This improved the situation and could locate the left root somewhat reliably and occasionally found the right root but they both commonly ran into trouble. The interface is made up of a collection of points but they are not evenly spaced, some are far closer than others which would result in incredibly high curvatures nowhere near the roots.

The approach was then changed to be simulation specific and to use knowledge of the geometry of the desired roots themselves rather than calculating the curvature. For the drop in a pool example, the stopping criteria used for the left and right roots acted as good approximations for these locations, so the left root came to be defined as the lowest point before the jet began to rise and the right root as the furthest point to the left before it went back on itself again. The former proved to be a great definition so was kept for the duration of the project, the latter was a decent approximation for the early formation of the jet but as it formed further the root would be placed halfway up the jet. In Figure 4, for example, it would land around $y = 2.65$ rather than $y = 2.55$. Thus, the definition was changed to be the point at which the gradient between that point and the next went below the value of -1 . This definition proved to work well and was its final definition. For the simulation with two droplets splashing, it was the points closest to the y -axis below and above the outer bulges of the droplets themselves, ignoring the air bubbles which form in the centre of the splash. In Figure 4 that means below and above $y = 3.0$ and $y = 2.0$.

With this approach, other features of the splash would commonly interfere with the nearest neighbour search and cause premature termination. With the right root, for example, ejected droplets would land back on the surface of the water and the search would fail. For the left root the jet would hit the droplet in some places and terminate the search early, such as in Figure 5. At the early formation of the splash, however, the left root could be found accurately, so the search for the right root would begin based on the location of the left root. It was found that beginning the search with the y -coordinate 3% lower and x^* 5% further out gave robust results. From that starting point it would traverse towards the y -axis until the above stopping

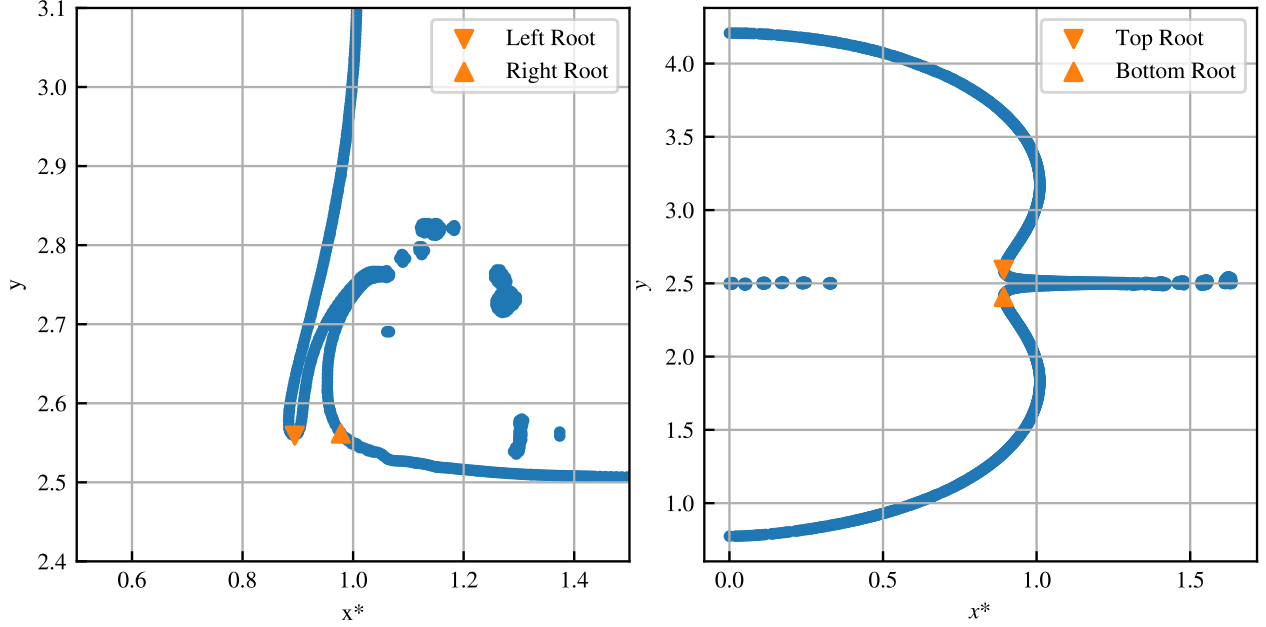


Figure 4: The locations of the roots of the jet in the case of a droplet landing in a pool (left) and two droplets colliding (right).

criterion was met. For the simulations of two droplets colliding, the simulation was approximately symmetric about the point $y = 2.5$, so if the top root was found to be at point $\mathbf{x}_0 = (x_0, y_0, z_0)$, the search for the bottom root would begin at the point $(x_0, 5 - y_0, z_0)$.

At the early jet formation stage this approach was strong, but still around 1 in every 20 roots would be misplaced, and for later portions of the simulation the start point for the right root needed to be further from the left. To address these, knowledge of the physics involved was used, namely that the jet region should be continuous in both space and time. To ensure time continuity, given the location of the roots from the previous frame, the code would begin the search from that point. This meant finding the four nearest points to each root and, in the left case, choosing the point with the lowest y -coordinate, and for the right case, ordering them by y -coordinate and finding the gradients between each until either the gradient went below a value of -1 or the final point was reached. Below is some pseudocode to demonstrate this algorithm for the droplet impacting a pool.

```
def find_roots(data, prev_known, prev_left_root, prev_right_root):
    if prev_known == True:
        n_points = four_nearest_neighbours(prev_left_root)
        left_root = point in n_points with lowest y value

        n_points = four_nearest_neighbours(prev_right_root)
        order n_points by y-coordinate
        m = gradient between n_points[0] and n_points[1]
        i = 0
        while m > -1 and i < 2:
            i += 1
            m = gradient between n_points[i] and n_points[i+1]
        if m > -1:
            right_root = n_points[i]
        else:
            right_root = n_points[3]
    if prev_known == False:
```

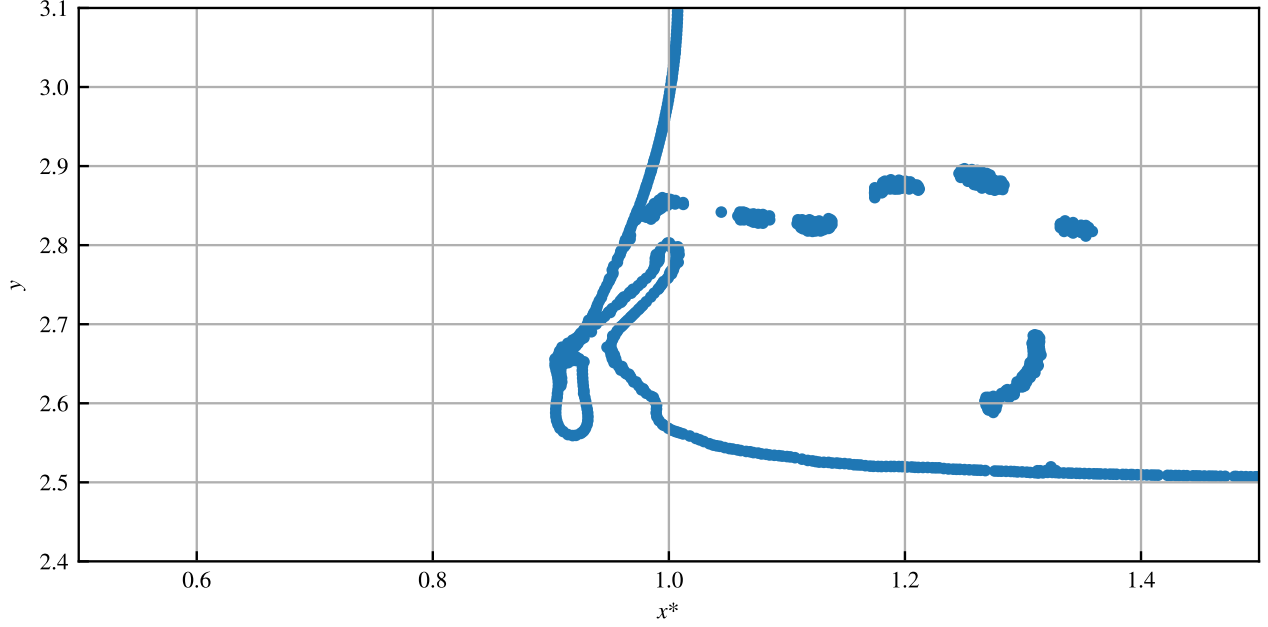


Figure 5: The Jet hitting the droplet making a root search difficult.

```

Begin search at top coordinate
Traverse down using nearest neighbour searches
Find last point before next has greater y-coordinate
left_root = last coordinate

start_point = 0.97*left_root
start_point[1] *= 1.05
Traverse left using nearest neighbour searches
Find last point before gradient < -1
right_root = last coordinate
return left_root, right_root

```

Testing found both roots were found well, for the right root the fourth point was rarely reached before the gradient reached the set criterion meaning that the root commonly lay within this search range. Similar results were found in the droplet collision simulations.

In order to fix space continuity, knowledge of the geometry of the splash was used. Since the droplets are spherical, they produce elliptical splash regions. Therefore, once all roots are approximated within a single frame, an ellipse of best fit is found amongst the points using a numerically stable version of direct least squares fitting by Halir & Flusser [8] which improves upon the method by Fitzgibbon et al. [7], the code used was taken from the Learning Scientific Programming with Python website [9]. This code finds the formula for an ellipse in the form

$$ax^2 + bxz + cz^2 + dx + ez + f = 0.$$

From there, the found roots are looped through and their x^* and y coordinates are substituted into the function

$$F(x, z) = ax^2 + bxz + cz^2 + dx + ez + f.$$

If $|F(x_i, z_i)| > \epsilon$, the root is moved onto the ellipse as follows: the root lies on a certain line of the x - z plane with a known angle, θ , this angle is converted into a gradient by

$$m = \tan(\theta),$$

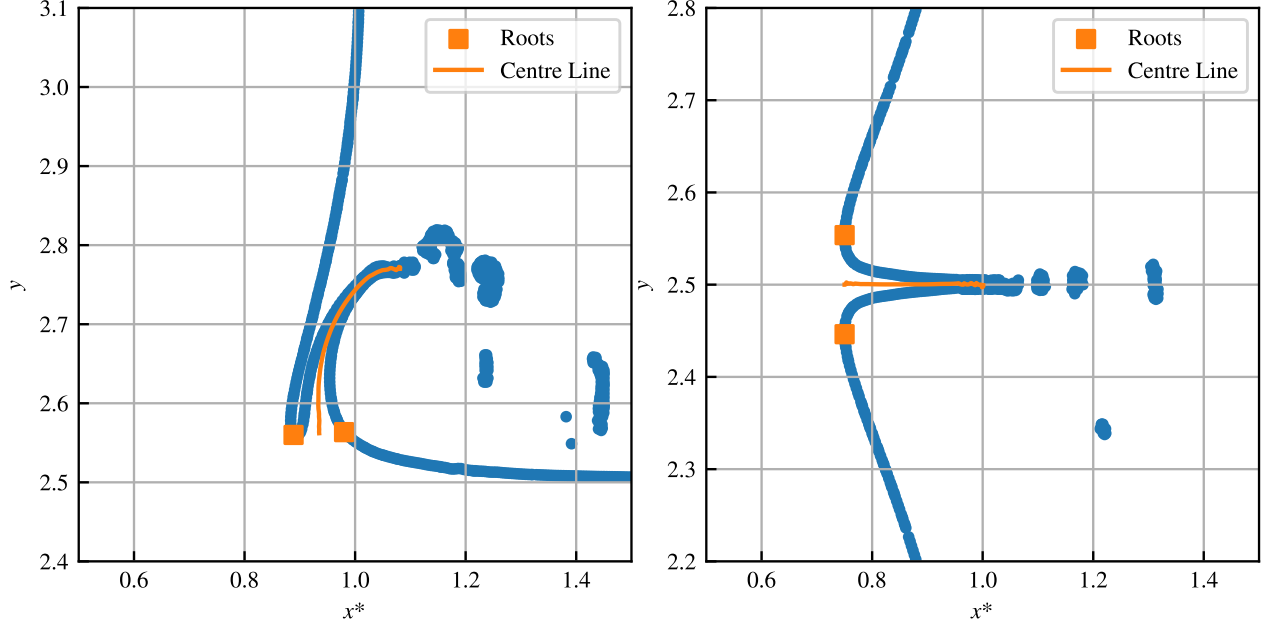


Figure 6: The centre lines of the jet in the case of a droplet landing in a pool (left) and two droplets colliding (right).

then the root lies on the line $z = mx$. This can be substituted into the formula as

$$F(x) = (a + bm + cm^2)x^2 + (d + em)x + f,$$

and we solve for $F(x) = 0$. This gives two roots, but since these simulations only contain $x \geq 0$, we simply take the positive root. The other root must have a negative x value since the ellipse will be (approximately) centred at the origin and $z = mx$ goes through the origin, so the other point will be on the other side of the x -axis. The y -coordinate is moved to be the median value of the other roots. In testing, it was found that $\epsilon = 2 \times 10^{-3}$ was a reasonable tolerance, moving points that were certainly in the wrong place but not points that were at or around the true root. Once all of these improvements had been made, the code would only relocate one or two roots out of the 360 found for most frames and never more than about 10% of roots in any frame, even ones with problematic features, reinforcing that the code itself was robust already and the correction mechanisms were not overbearing.

Theories on the rate of growth of the roots over time as well as the distances between the roots were derived by Cimpanu & Moore [4], those being that the distance of the roots from the y -axis grows with $O(t^{1/2})$ and, in the case of the droplet impacting a pool, the distance between the roots grows with $O(t^{3/2})$. These theories will both be tested in the next section.

3.2.3 Mapping the Centre Line of the Jet

The way in which the jet evolves over time can be summarised by the line running along its centre, thus the ability to automatically map its evolution over time is of great use. Examples of the centre lines of the jet in each case are displayed in figure 6.

In order to find this centre line, a nearest neighbour search is used again. Starting at each root, the points along each side of the jet are mapped sequentially with conditions to make sure it searches in the correct direction: for the left root in the drop on a pool this means searching away from the y -axis, for the right root this means searching upwards, for the top root of the two droplets colliding this means searching downwards and for the lower root searching upwards. Once a new point on both sides is found, the midpoint of these two points is appended to a list. The search continues until the nearest neighbour search of one of the sides finds a point previously searched by the other, it then terminates and returns a list of the midpoints and the points of the jet on each side. Finding this is a useful tool to find the thickness of the jet along its length.

3.2.4 The Thickness of the Jet

The final feature of the splash to analyse is the thickness of the jet through time and space. At the base of the jet in the droplet on a pool example, the left root can be very messy with many points bunched up together, and towards the tip of the jet, as factors such as surface tension cause the jet to break up and form droplets, similar occurs. This means that the points found in these areas can give misleading values and are less relevant for analysis, therefore only the parts of the jet between 20% and 80% of its length are used for the thickness analysis.

Two different approaches were taken to find the jet thickness, one being better for the droplet on a pool simulation, the other better for the droplet collision simulation. In both methods, the gradient, m_i at each point, $\mathbf{x}_i = (x_i^*, y_i)$ along the centre line is approximated by using constrained regression to find the line of best fit for the four points $\mathbf{x}_{i-2}, \mathbf{x}_{i-1}, \mathbf{x}_{i+1}, \mathbf{x}_{i+2}$ which pass through $\mathbf{x}_i$. This can be calculated as

$$m_i = \frac{\sum_{j=i-2}^{i+2} (x_j^* - x_i^*)(y_j - y_i)}{\sum_{j=i-2}^{i+2} (x_j^* - x_i^*)^2}.$$

From there, the first method finds the equation of the line which runs through $\mathbf{x}_i$ with gradient m_i . This has the equation

$$-m_i x^* + y + m_i x_i^* - y_i = 0.$$

Then, the two points which formed the said point from the centre line are found and the perpendicular distances to the line are calculated using the formula

$$\text{dist}_k = \frac{|Ax_k^* + By_k + C|}{\sqrt{A^2 + B^2}},$$

where $Ax + By + C = 0$ is the equation of the line and $\mathbf{x}_k = (x_k^*, y_k)$ is the point on the jet. The perpendicular distances from each of the points are then added together to approximate the thickness of the jet.

The other method instead finds the equation for the normal to the centre line at the point in question, if the gradient is incredibly close to 0 then the gradient of the normal is set to a value of 100. The perpendicular distances to every point on the jet are then found in the same manner as above and saved in a list. The two points with the closest perpendicular distance to this normal are then found and the thickness of the jet is defined to be the distance between these two points. The code ensures that they are on opposite sides of the jet by checking whether the distance between the two points is smaller than the distance to the centre line, if it is greater then the search continues until this condition is satisfied.

Finally, the thickness values are smoothed to guard against outliers. The values in the middle of the jet are the most robust, so the midpoint of the centre line is taken, then the code iterates along the jet in both directions, if the thickness of an adjacent point is greater than 10% of the previous then it interpolates it by taking the average of the thicknesses for both neighbouring points. If it is a thickness from the end of the jet, it is interpolated to have the same percentage difference in thickness from the penultimate point as the penultimate point had to the antepenultimate point. In testing, there were rarely more than two thickness values interpolated.

A visual representation of the algorithms in both studied cases is shown in figure 7. The top left shows the first approach for the jet formed from a drop hitting a pool, we see that at the bottom some of the lines spill over the left hand side of the jet and are not properly centred whilst at the tip they are too far over in the other direction. This is because the two points which make any specific centre line may not be directly opposite each other along the jet and are commonly diagonal in this example. The top right shows that the second approach is superior for this form of jet since none of the lines spill outside of the jet. The bottom right shows the second approach for the jet formed from two droplets colliding. Due to divisions by zero from the nearest points on either side of the jet having the same x^* value, not many of these lines would even render and of the ones which did, some are quite diagonal and not perpendicular to the centre line. The bottom left shows that the first approach is better for this type of jet since all the lines are perpendicular to the jet and none are spilling outside of the jet region.

From discussions with my supervisor, Radu Cimpanu, along with others within the research group and wider computational fluids community, it is anticipated that in the case of the two droplets colliding, the thickness of the jet shrinks with order x^{*-7} along the length of the jet. This will be tested in the next section.

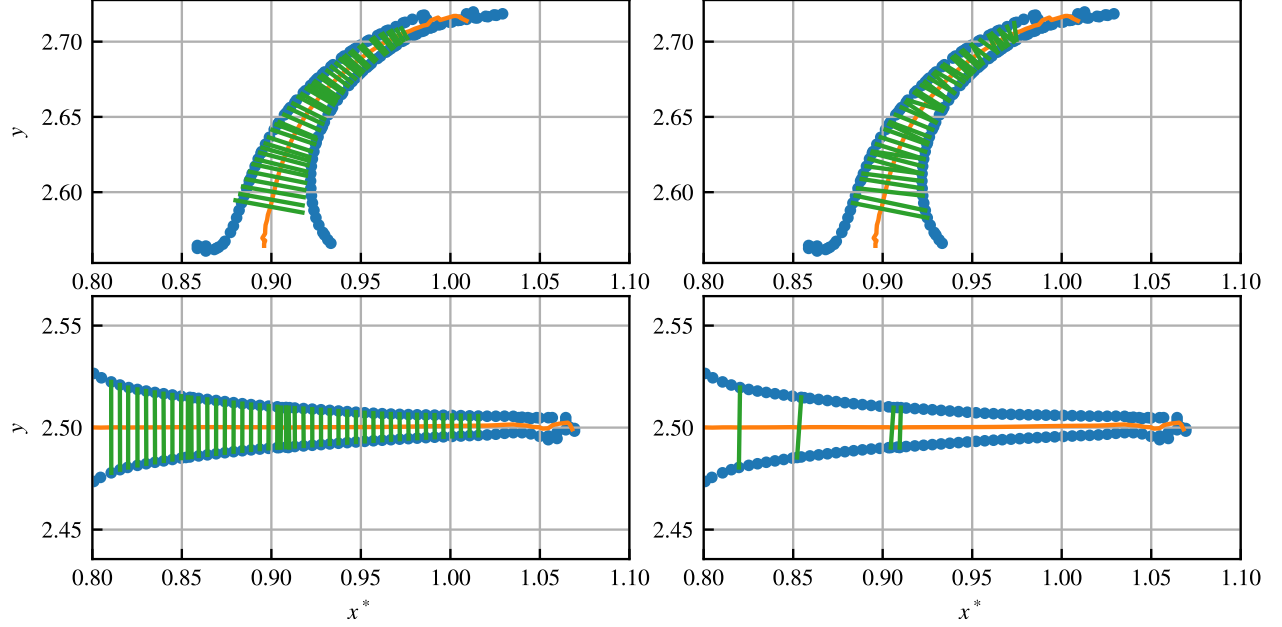


Figure 7: Top row: jet formed from a drop onto a pool. Bottom row: jet formed from two colliding droplets. Left column: first described approach. Right column: second described approach.

4 Results and Discussion

4.1 Ejected Droplet Analysis

The analysis in this section comes from four different datasets: a lower and higher resolution simulation of both a droplet landing on a pool and two droplets colliding with each other.

The first statistic to measure was a simple droplet count over time. Since only the ejected droplets are of interest, only bodies with a volume value of less than 2.0 are considered (the original droplet had a volume of 2.1). In figure 8, we see a count of the number of ejected droplets over time for a droplet landing on a pool. It is clear that the lower resolution simulation is not powerful enough to draw any meaningful insight from the ejected droplets since it only measures at most six ejected droplets when in the higher resolution it is clear there are far more. This is because most of the ejected droplets are miniscule and at a low resolution they do not render. In fact, in the higher resolution the first ejected droplets are detected at around time $t = 0.35$ but in the lower resolution none appear until nearly $t = 0.45$. Even the higher resolution has some issues in droplet detection, around a time of $t = 0.4$ some droplets begin to disappear, by inspecting the raw data this is not due to the droplets recombining with the larger body of fluid, but because they go out of range and the software loses track of them. For this reason, subsequent analysis will focus only on the higher resolution simulation.

It is clear that the rate of droplet production increases as time passes, at least for as long as the simulations lasted. We can know this since droplets ejected at a high velocity earlier in the simulation de-render when they get too far from the splash zone but the number of droplets still increased over time in the higher resolution simulation. For the ejected droplet count to rise whilst early droplets de-render must then mean that the rate of droplet production increases as time goes on. At some point the droplet ejection will hit a peak but since this simulation only continues for approximately around $20 \mu\text{s}$ after collision this point is not reached or if it has then there is no way of telling. The simulation stopped earlier than originally anticipated due to a solver issue potentially caused by a small bubble or droplet causing pressure build up, thus if this could be fixed and the solver run for longer and a better view of the droplet behaviour could be inferred. Wang et al. [28] found the peak number of ejected droplets to be around $20 \mu\text{s}$ which is incidentally approximately how long the simulation lasted after collision.

Figure 9 shows the same statistic for the simulations with two colliding droplets and the software manages

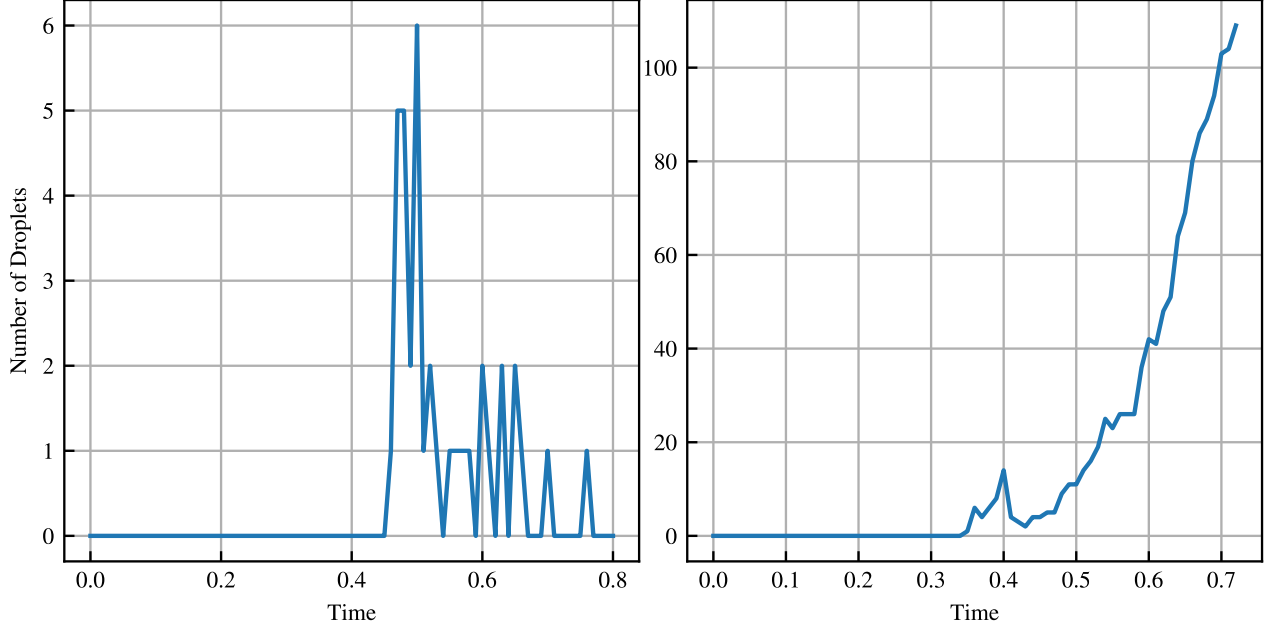


Figure 8: Number of ejected droplets over time for droplet on pool simulations. Left: lower resolution simulation. Right: higher resolution simulation.

to pick up far more of the droplets. Where the lower render found at most seven ejected droplets in the previous example, this managed up to 58. More were found in the higher render too with a maximum of over 175 compared to around 110 previously. In both simulations the number of droplets begins by increasing once the collisions occur and rise until around $t = 0.6$ when the number starts falling. This occurs because many of the droplets that were released earlier go out of frame and the code no longer renders them. Checking the data by hand shows that by the final frame the total volume in all the identified droplets has dropped by around 2.9×10^{-3} from an original total of 64.5. This is a minor change in the grand scheme of the simulation, but proves that some droplets really do disappear.

The peak droplet ejection rate is found and appears to be at around $t = 0.57$ in both the low and high resolution simulations. At this point in the simulation, the colliding droplets have fully merged and the roots of the jet somewhat cease to exist, there is no longer a turnover point such as the one shown in figure 4, now as the interface is traversed the points all move further away from the y -axis and the droplet does not go back on itself. This change in geometry may be what causes the ejection rate of the secondary droplets to decrease.

The next statistics to analyse were the average speed of the ejected droplets over time. This statistic for the high resolution simulation of a drop hitting a pool of water (again removing any bodies with a volume less than 2.0) is displayed in figure 10 and it shows a clear decline over time. There is also a dip in the average speed at $t = 0.4$ which corresponds to the sudden up tick in the number of droplets seen in 8 before the average speed increases again and the number of drops decreases. This may seem like a negative correlation exists between the number of droplets and their average speed but the other simulations imply this is not the case, what we really see is a positive correlation between the volume of droplets ejected and their speeds.

Figure 11 shows the average speed of the ejected droplets over time for both the low and high resolution simulations and we again see a general decline over time. If it were true that there was a negative correlation between droplet count and average speed then both of these graphs would begin to rise after $t = 0.6$ since this is when the droplet counts fall in Figure 9. This, however, is not what we see so something else is occurring. What we notice is that at every time step where data is available for both simulations, the average speed from the lower resolution simulation is higher than that of the higher resolution simulation. Since the higher resolution simulation captures droplets with lower volumes, this implies that in fact there is a positive correlation between a droplet's volume and its speed. Figure 12 shows a scatter plot of the droplet speeds

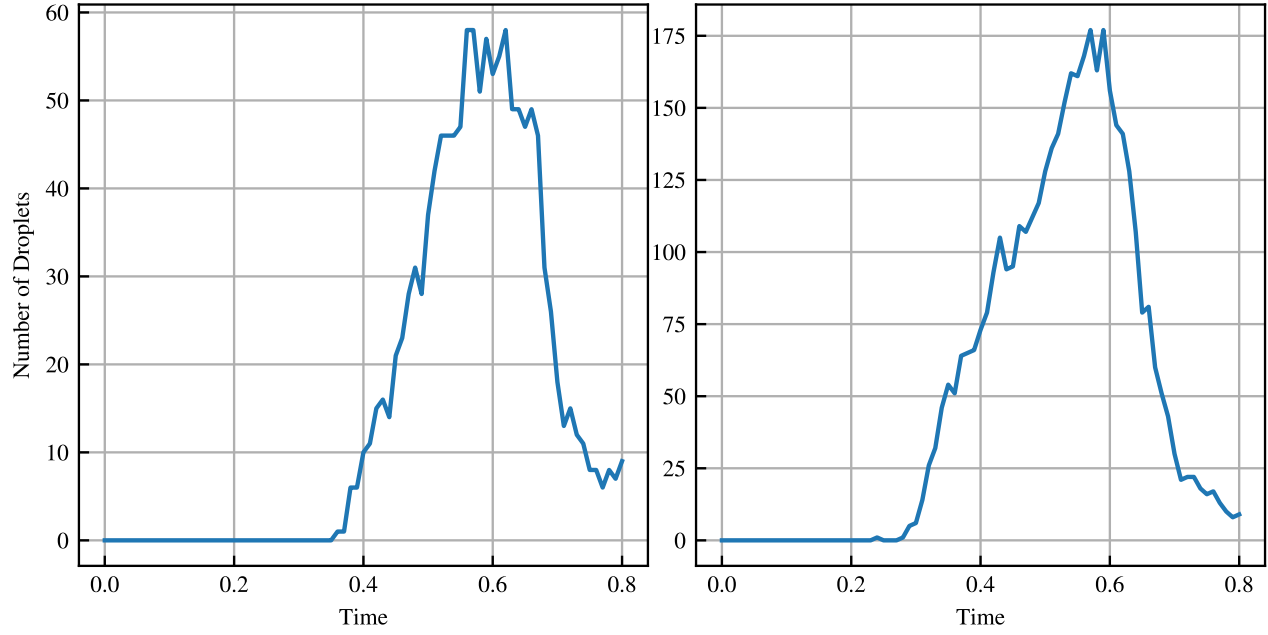


Figure 9: Number of ejected droplets over time for simulations of two droplets colliding. Left: lower resolution simulation. Right: higher resolution simulation.

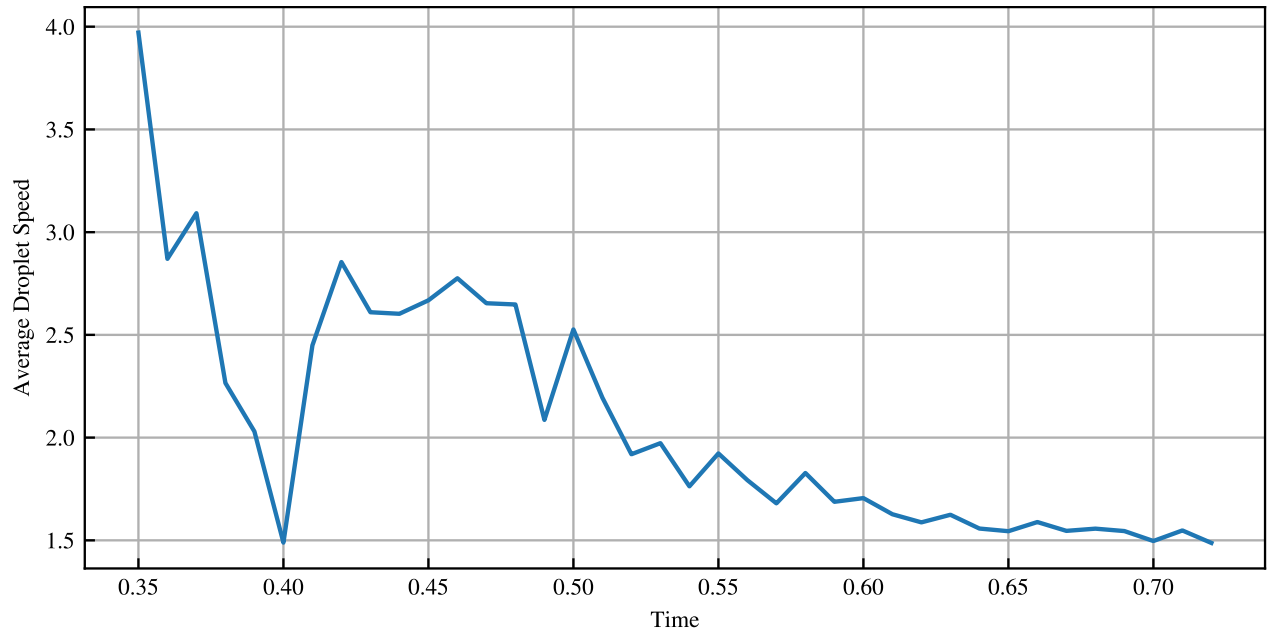


Figure 10: Average speed of the ejected droplets over time from a simulation of a droplet on a pool.

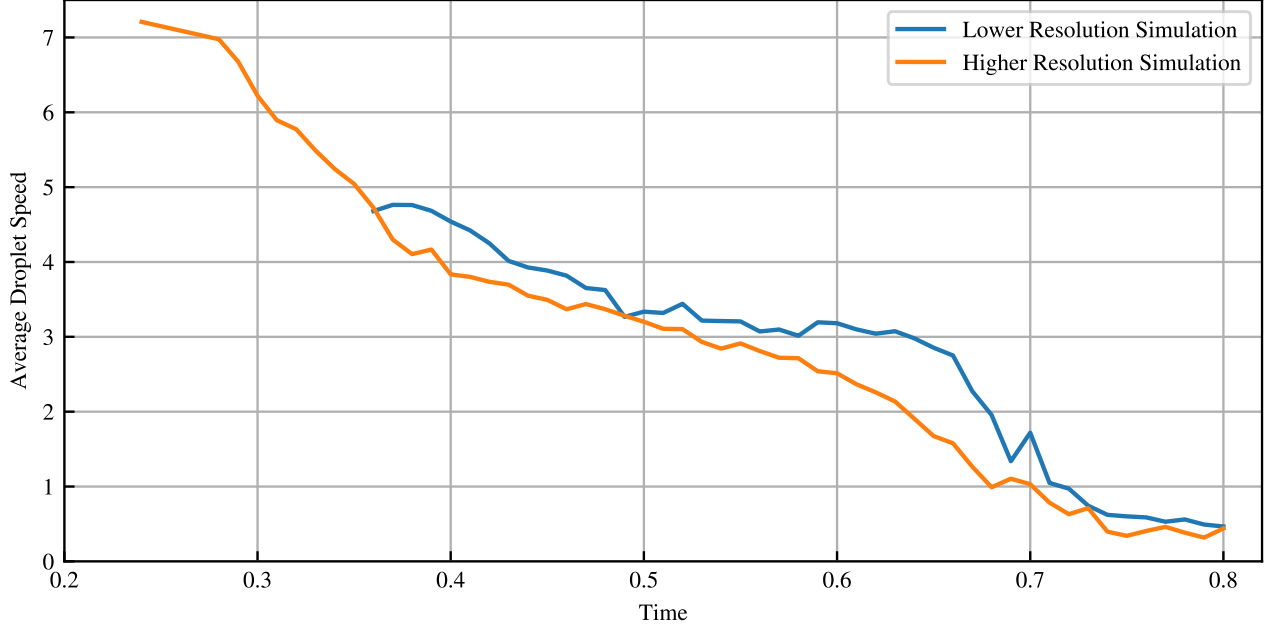


Figure 11: Average speed of the ejected droplets over time from a simulation of two droplets colliding.

against the log of the droplet volumes at times $t = 0.4$ and $t = 0.59$. There is a clear positive correlation in both graphs, ignoring the few outliers of droplets which have miniscule volume, but also the speed of all droplets appear to generally decrease.

The velocity behaviour being a general decrease over time is in agreement with both theoretical results and other numerical results by Burzynski et al. [2] and Wang et al. [28]. This happens because the droplets are ejected from the tip of the jet when surface tension factors cause it to break up, as the jet gets larger the velocity closer to the tip slows down, therefore later droplets having slower speeds is expected. Moreover, the droplet speeds being positively correlated with their volumes within the same time frame was also replicated by Burzynski et al. [2], solidifying both the theory and the simulation's reliability.

The final piece of analysis for this section is to look at the 3-dimensional histograms for the droplet volumes over time. For this, only the high resolution plots will be used since they give the best view of what happens. The time steps considered have been reduced to only show those with multiple droplets since otherwise the density plots become unreadable for later time steps when the spread increases.

The density histogram plot for the simulation of a droplet hitting a pool is displayed in figure 13. We see that early on in the splash, the density is strongly concentrated around a volume of 1×10^{-6} before spreading out more later on. The frequency plot in figure 14 reveals that this is due to a very small number of droplets being rendered at that point, it is not until later on that more droplets form and a distribution begins to form. By $t = 0.7$ the distribution appears to be centred around a volume of 1×10^{-5} .

Figures 15 and 16 show the outcomes of the same analysis for the simulation of two droplets colliding. This time there are more droplets at the earlier portion of the splash and the density is instead more tightly concentrated at the later time steps. The frequency plot reveals that this is again due to the low number of droplets at this point in time and that rather most droplets are found between times $t = 0.4$ and 0.6 . At the times with the most droplets, they are somewhat evenly spread from 10^{-5} to 10^{-6} before concentrating around a volume of 10^{-6} . In both simulations, the minimum and maximum volumes of the secondary droplets are 10^{-10} and just over 10^{-4} .

Comparing this to the literature, we see close agreement with the analysis from Burzynski et al. [2]. Although that paper concerns droplets splashing onto solid surfaces, the early time droplet splashing theory still applies to splashes onto liquids so comparison is possible. In figure 11 from that paper, we see graphs for the distribution of droplet radius sizes as a percentage of the original droplet's radius. The distribution of these droplet radii are shown in panel (a) of this figure, we see the majority of the droplets lying within a

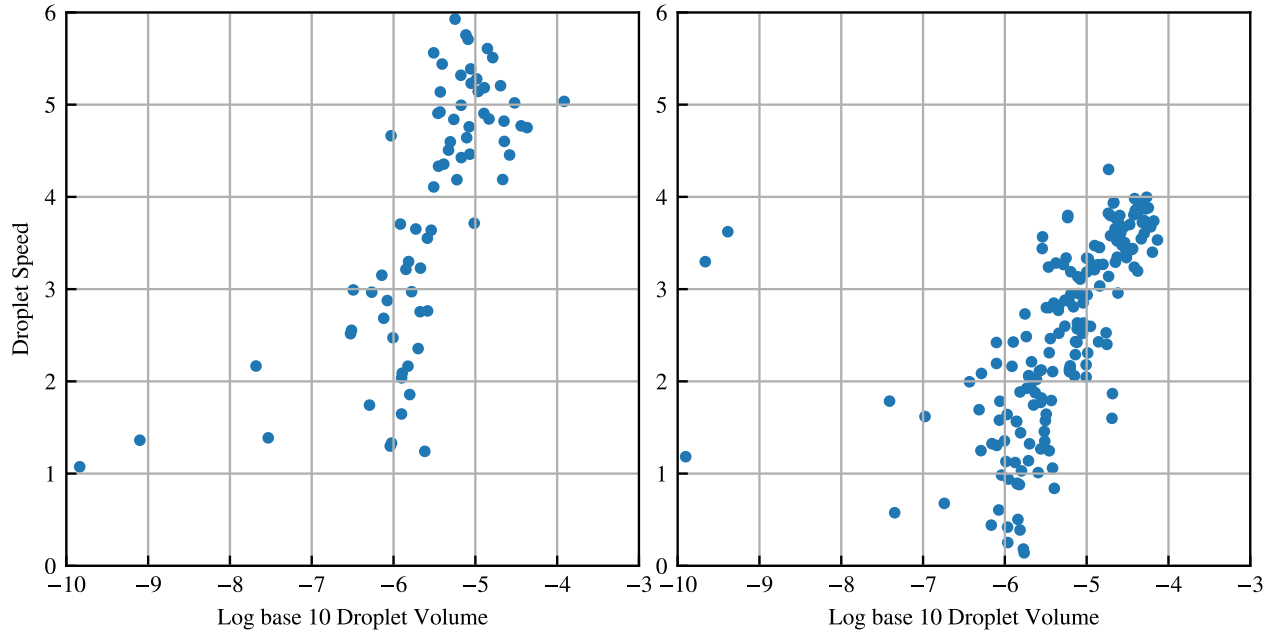


Figure 12: Droplet speeds against log volumes at times $t = 0.4$ s (left) and $t = 0.59$ s (right).

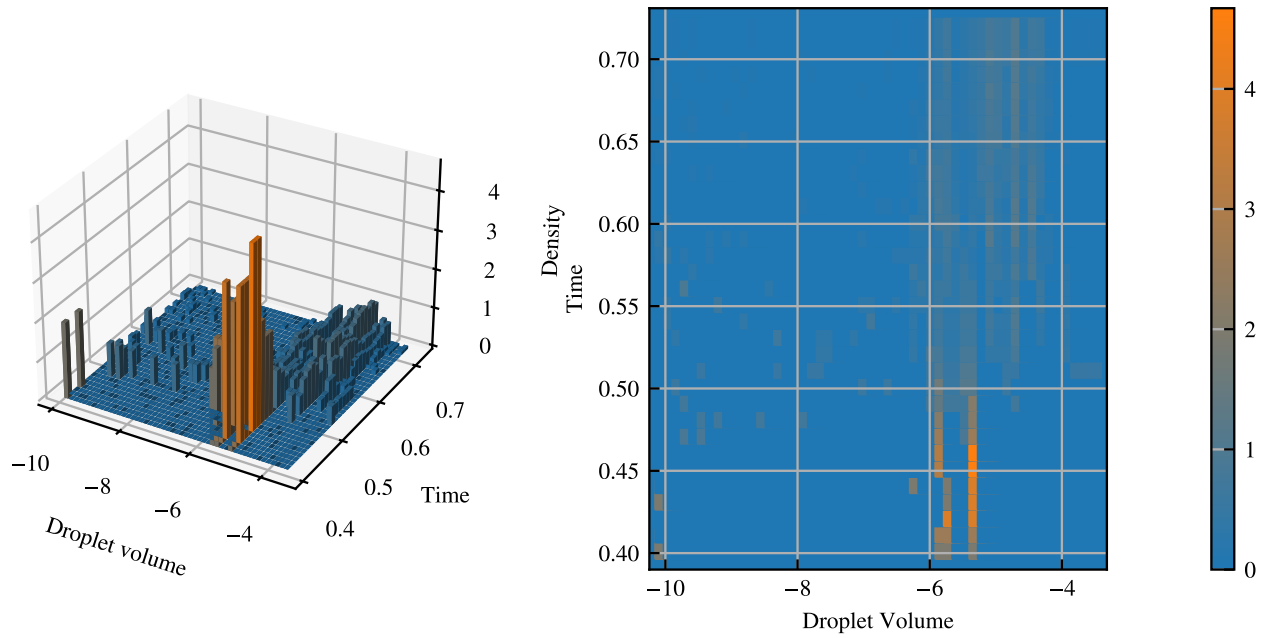


Figure 13: Three-dimensional density histogram of log droplet volumes against time (left) and the corresponding contour plot of the densities (right) for a droplet splashing on a pool.

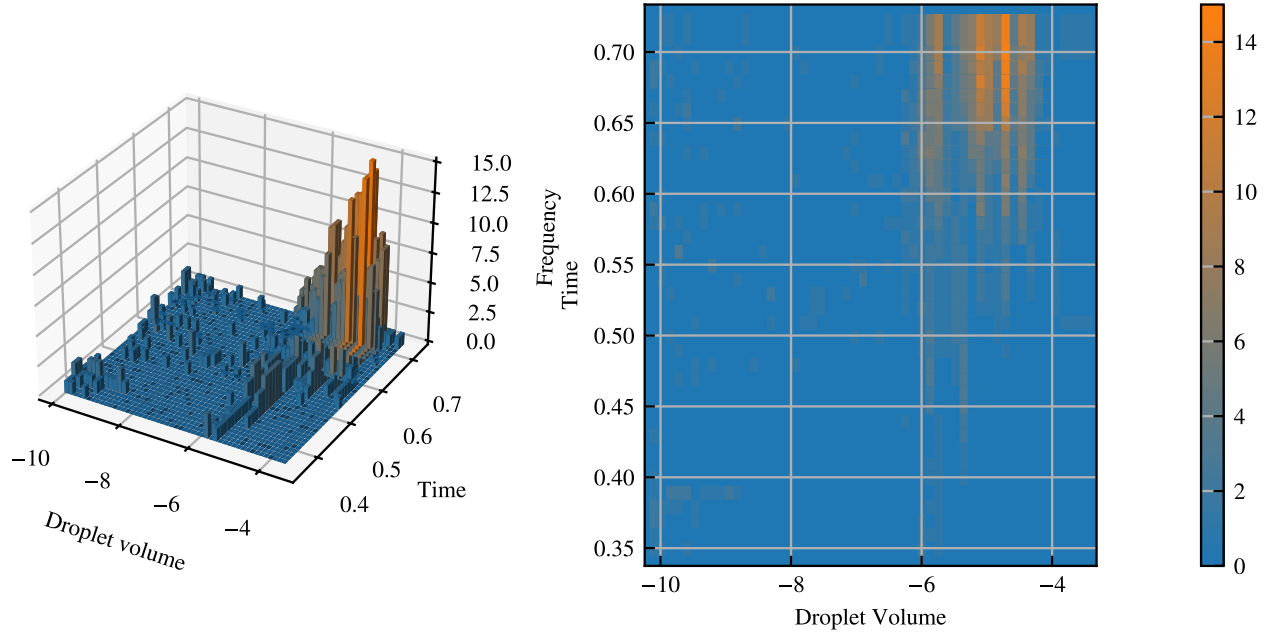


Figure 14: Three-dimensional frequency histogram of log droplet volumes against time (left) and the corresponding contour plot of the densities (right) for a droplet splashing on a pool.

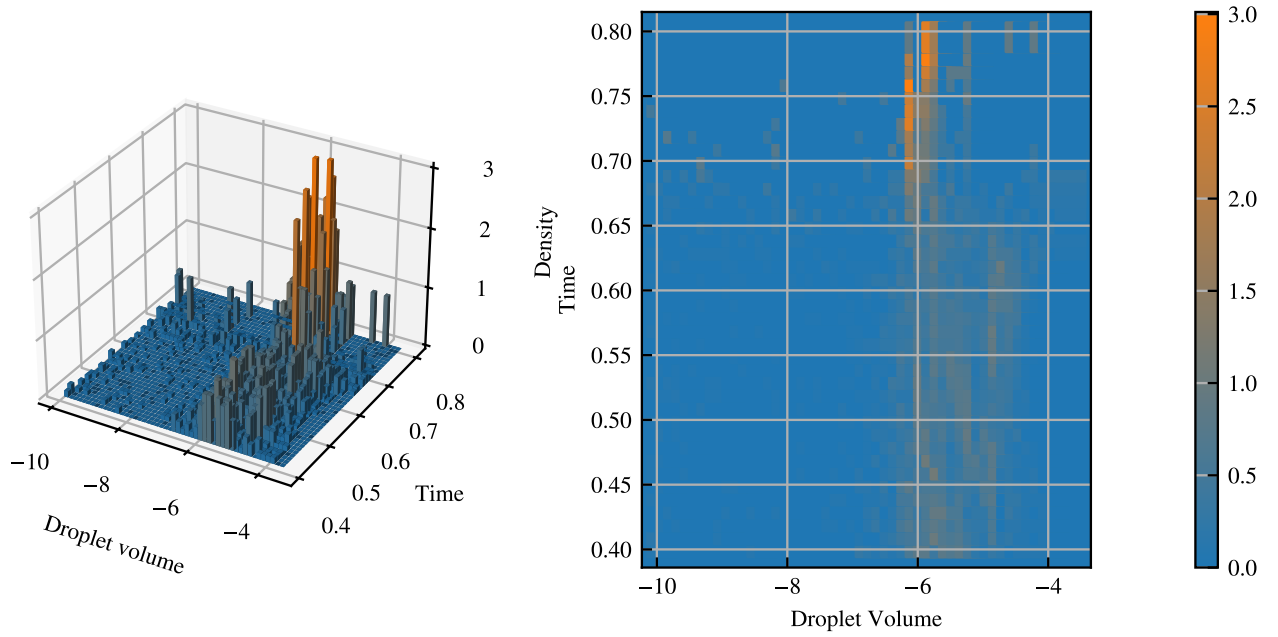


Figure 15: Three-dimensional density histogram of log droplet volumes against time (left) and the corresponding contour plot of the densities (right) for two colliding droplets.

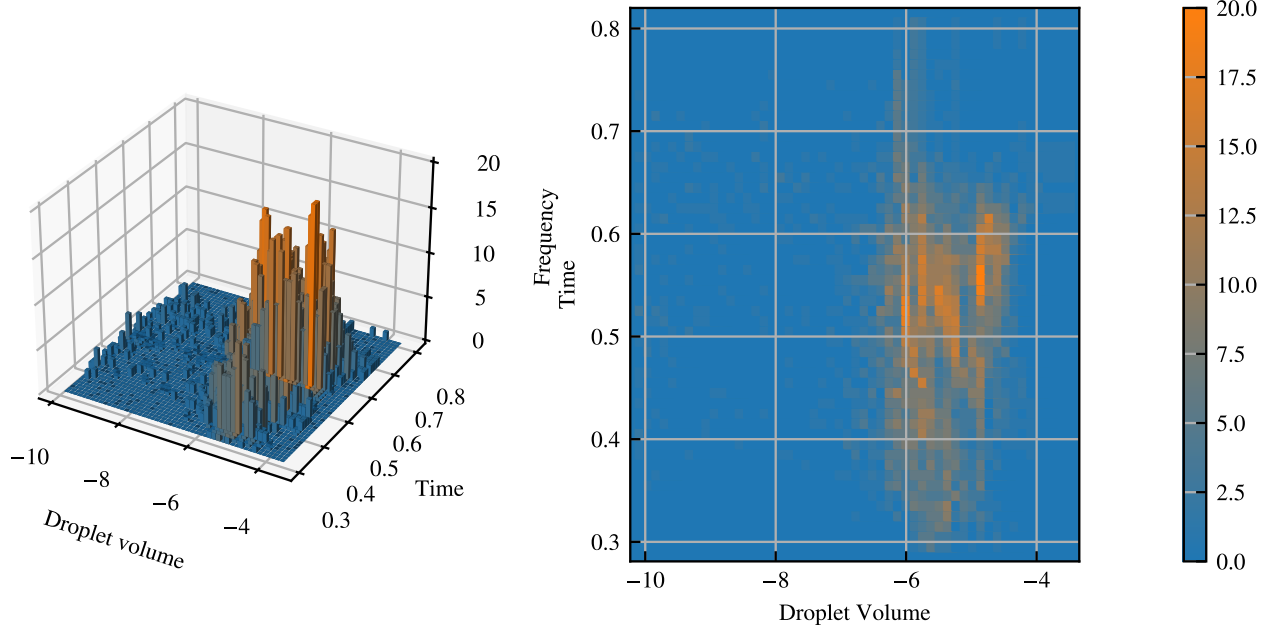


Figure 16: Three-dimensional frequency histogram of log droplet volumes against time (left) and the corresponding contour plot of the densities (right) for two colliding droplets.

range of 0.5-1% of the original droplet radius with the smallest droplets being around 0.2% and the largest being 3%. In order to compare this to our analysis, we have to convert the volumes to radii and divide this value by the original value. The original volume was around 2.0 so for any droplet volume V we simply perform

$$d/D\% = 100\sqrt[3]{\frac{V}{2}}.$$

Our quantities of interest were volumes of 10^{-4} for the upper bound, 10^{-5} - 10^{-6} for the mean values and 10^{-10} for the smallest droplets. Converting this to the radius percentage, we get values of 3.68%, 1.7%, 0.79% and 0.0368% respectively. The largest droplets are then only slightly larger than those observed by Burzynski et al., and the smallest within the observed range and larger than the theoretical minimum proposed by Cimpeanu & Papageorgiou [5]. Moreover, the values where most droplets lie also agree with the spread shown in the paper by Burzynski et al.. The smallest droplets from the simulation are at the very edge of the possible volumes that can render, to test whether these really are the smallest possible droplets a simulation with a finer rendering would need to be performed though this would take months on multiple CPUs with the current software. Further reasoning for increasing the refinement level is that the referenced analysis by Wang et al. identified thousands of ejected droplets at its peak whilst this simulation found less than 200, another level of refinement may make the analysis more insightful. Overall, there is strong agreement with both theory and previous analysis gives a strong case to support that simulations of this refinement level via the BASILISK [19] software are reliable approximations for the real life phenomena being simulated, justifying the computational cost of producing such simulations.

4.2 Interface Analysis

For this part of the project, five datasets were available: a high resolution simulation of a droplet landing on a pool which the algorithms were originally created on, and four datasets corresponding to those discussed in the previous analysis. Unfortunately, the lower resolution simulation of the drop landing on a pool was not refined enough to perform analysis to find the roots so is not used here. Therefore, there are two simulations of droplets hitting a pool of water, these are both very similar in resolution levels but the simulation studied in the previous section ran for an extra few frames so that will be the one used for the most part, though plots from the other simulation may be used since they are practically interchangeable. The higher and lower resolution simulations for the droplet collision produced very similar results for the locations of the jet roots, but for the lower resolution simulation the jet did not form as well in many cases making the analysis for its centre line and thickness less reliable. Due to this, the results from each analysis tool will only be shown twice: once for each type of simulation coming from its higher resolution version.

4.2.1 Jet Root Evolution

As the simulations progress, the locations of the roots are not stationary, they move over time. Figure 17 shows how they move away from the origin over time, the rate at which this happens is of particular interest. The graph shows the median distance of the roots from the y -axis, since the splash region is circular in both simulations they have been labelled as the radii. We see that the distance of a root, r , from the origin seems to be approximately proportional to the square root of the time, $\sqrt{t}$ plus some constant, k . That is,

$$r \propto \sqrt{t} + k \text{ (approximately).}$$

The correlation coefficient between $\sqrt{t}$ and r is equal to 0.9976 for the left roots, 0.9988 for the right roots from the droplet on a pool simulation. The top roots from the droplet collision simulation give a correlation coefficient of 0.9929, strongly supporting the growth being $O(\sqrt{t})$. This behaviour was derived and simulated in two dimensions by Cimpeanu & Moore [4] and these results back this theory further. The regression lines are not perfect, however, and there are gaps at both ends between the measured lines and the regression lines. This implies that whilst the leading order term is likely to be $\sqrt{t}$, there are strongly influential lower order terms too. An animation of the root evolution over time for the droplet on a pool is available in the additional materials [14].

4.2.2 Jet Thickness and Centre Line Evolution

The final piece of analysis was to see how the centre lines and thicknesses of the jets evolve over time. Figure 19 shows how the centre line changes as time increases, we see that the jet grows over time and moves further away from the origin, as we saw with the roots, but its angle also changes. In its early formation we see that it is fairly flat but by the end the body of the jet begins to go back on itself before reaching out. The equivalent plot for the colliding droplets simulation has not been included since it does not reveal much. The jet in that case also moves away from the origin over time, as we saw with the roots, and the jet grows but the shape of its centre line does not change, it simply remains a horizontal line at $y = 2.5$. An animation to show the evolution of the centre line over time for a droplet on a pool is available in the additional materials [14].

Figure 20 shows that the distance between the roots of the jet grow over time at a rate proportional to $t^{3/2}$ for the simulation of a droplet on a pool. The regression line has a correlation coefficient of 1 (4dp) matching practically perfectly. This was also theorised by Cimpeanu & Moore [4] and the graph strongly supports this to be the leading order term in this case. For the case of two droplets colliding, growth of order $t^{3/2}$ is not seen but rather t^3 with a correlation coefficient again being equal to 1 (4dp), this result was not found in literature, this is simply what was found by playing with other potential regression lines. If not already present in other literature, this could be motivation for further analytic study to verify whether growth of this order is anticipated mathematically for this specific case.

Figure 21 shows how the thickness of the jet evolves along the jet through time. We can see that the base of the jet grows in thickness over time whilst the tip stays approximately similar, the jet also moves outwards in space for both cases as time progresses. It is anticipated by researchers in this area that the evolution of thickness along the length of the jet grows approximately with order $(x^*)^{-7}$. In figure 22, regression lines

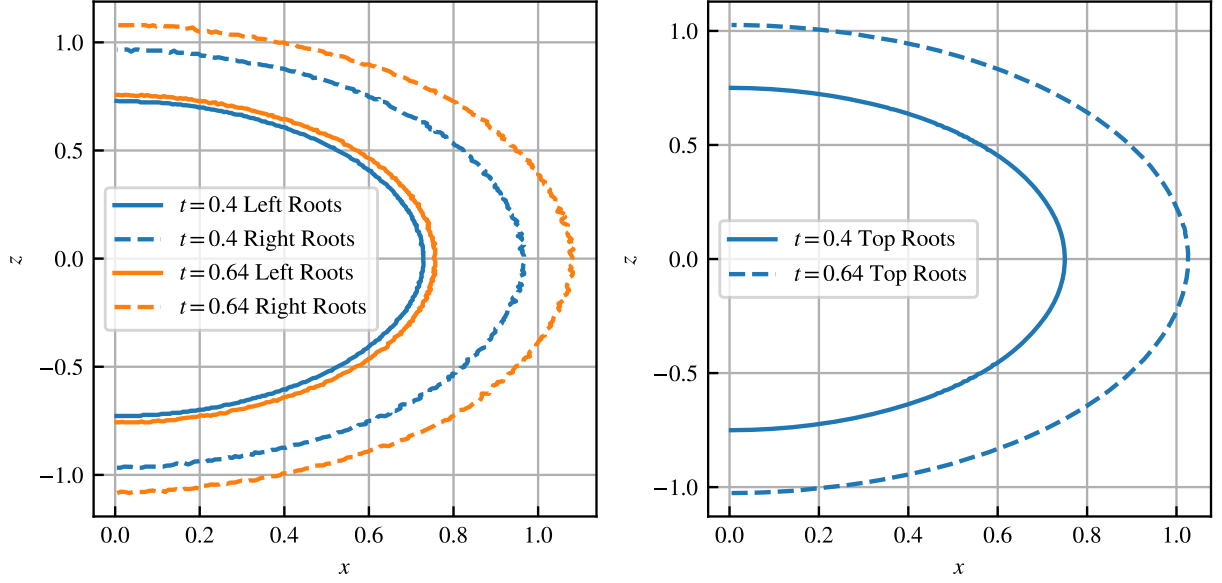


Figure 17: Jet roots move outwards over time. The left plot shows the left and right roots from the droplet on a pool simulation. The right plot shows the top root from the droplet collision simulation.

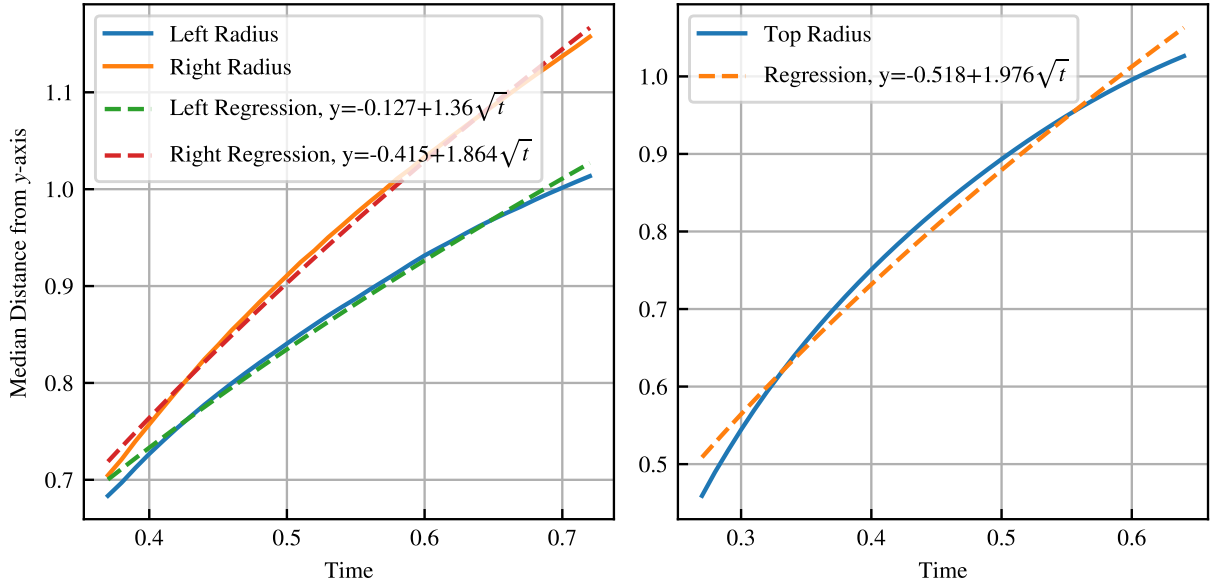


Figure 18: Regression lines for the median distance of the jet roots from the y -axis. Left: droplet on a pool simulation. Right: droplet collision simulation (the top and bottom roots are almost precisely the same so only the top root is shown).

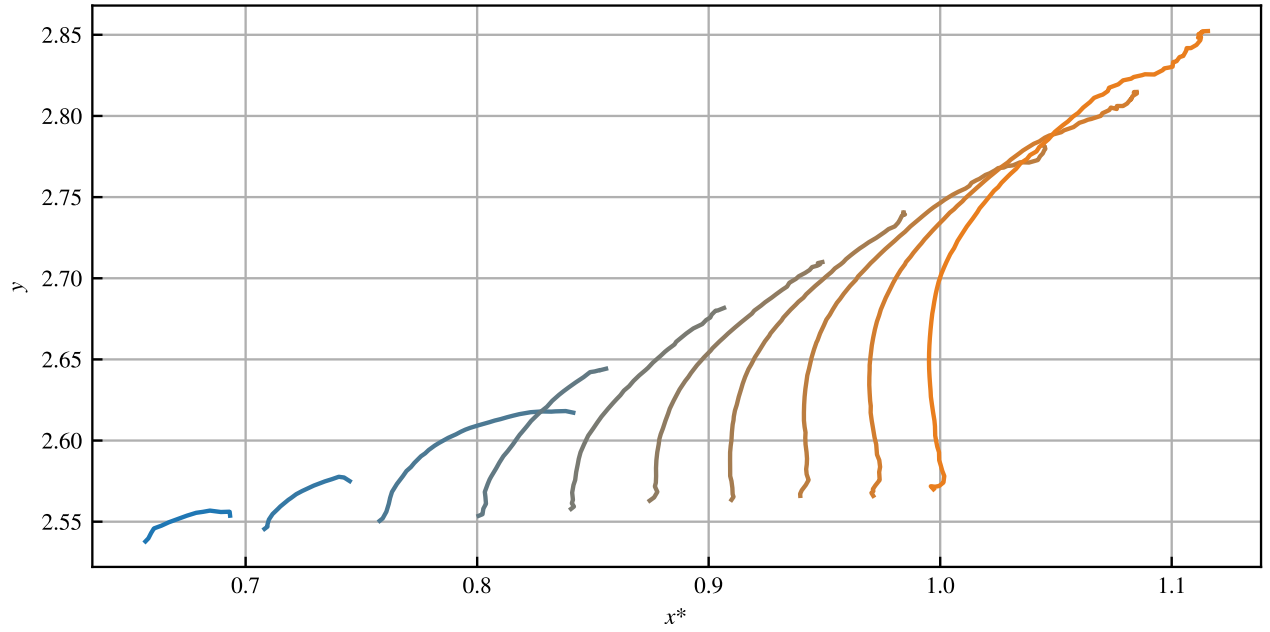


Figure 19: The evolution of the centre line through the jet as time increases. The line furthest to the left is the earliest jet at time $t = 0.35$ s and the furthest to the right is time $t = 0.62$ s with increments of 0.03 s between each subsequent line.

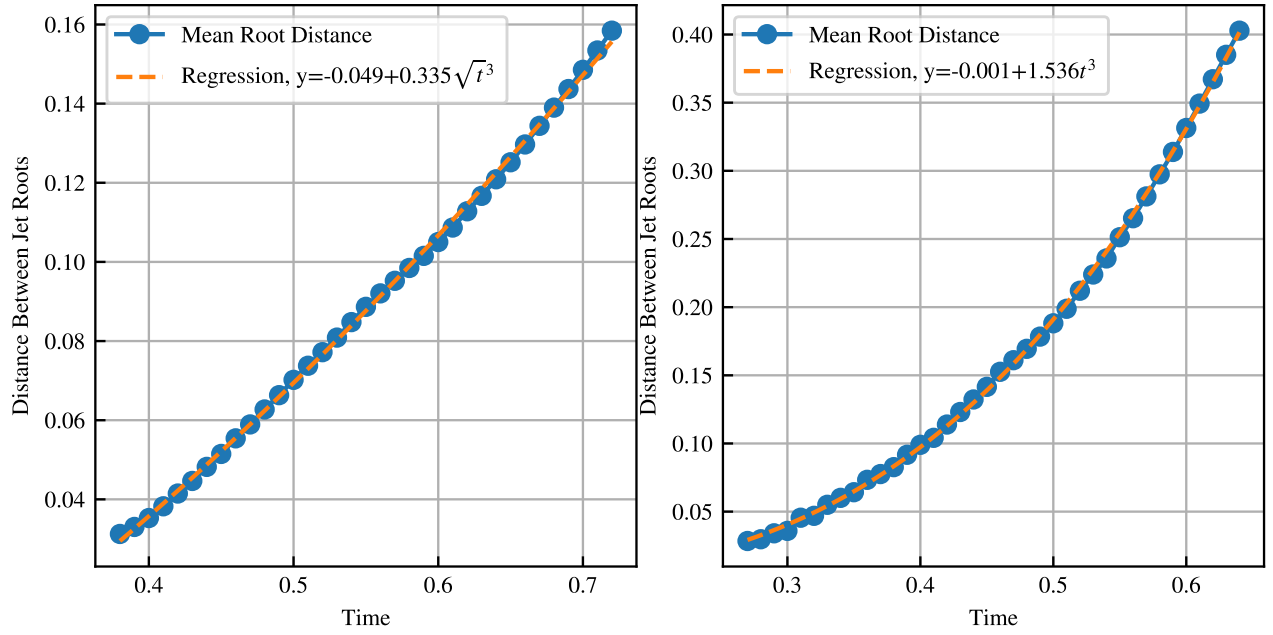


Figure 20: The change in the distance between the roots of the jet with relevant regression lines. Left: simulation of a droplet on a pool. Right: simulation of two droplets colliding.

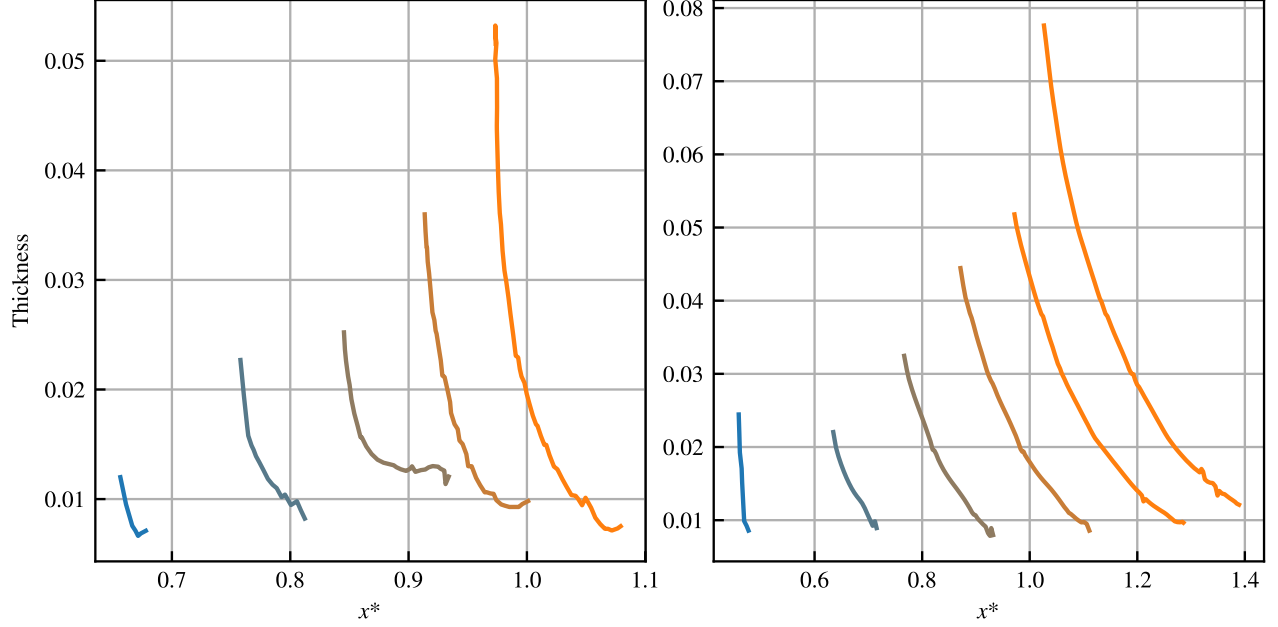


Figure 21: The evolution of the thickness of the jet as time increases. Left: The jet formed from the droplet on a pool simulation. The line furthest to the left is the earliest jet at time $t = 0.35$ and the furthest to the right is time $t = 0.62$ with increments of 0.06 between each subsequent line. Right: The jet formed from the droplet on a pool simulation. The line furthest to the left is the earliest jet at time $t = 0.27$ and the furthest to the right is time $t = 0.63$ with increments of 0.06 between each subsequent line.

for this order are shown and a good fit is found for the case of two colliding droplets, whilst a weaker but believable fit for the case of a droplet on a pool when $||x||$ is used instead of x^* . This supports for the theorised hypothesis and gives some motivation for a formal paper to be formed testing this analytically. A potential reason for the weaker correlation for the droplet on a pool is that the centre line of the jet goes back on itself especially at the later times, this means that multiple thickness values are given for the same x^* value so a regression line does not work as well. A more precise formulation of the suspected evolution of the jet thickness through time may give stronger results than that which was provided from current speculation.

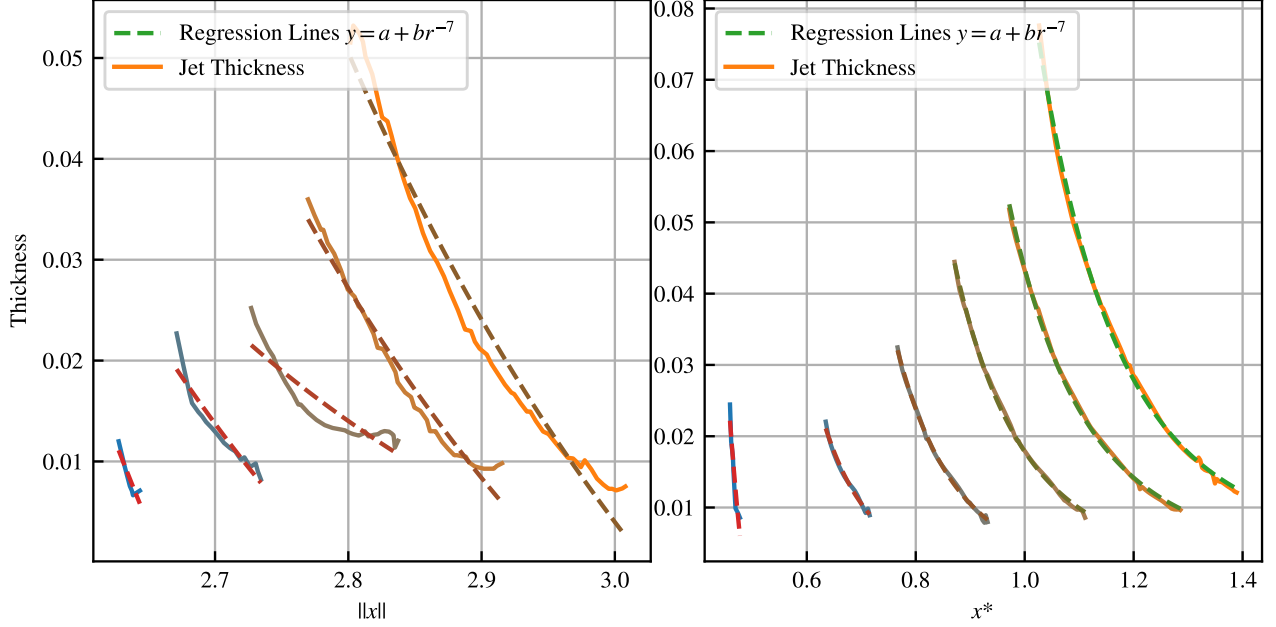


Figure 22: The evolution of the thickness of the jet as time increases with corresponding regression lines. Left: simulation of a droplet on a pool. Right: Two droplets colliding.

4.3 Impact

The results from the analysis carried out in this report have wide ranging implications and uses. The first is the evidence in support of multiple analytic theories and replication of previous results. Finding agreement with the theoretical and observed bounds on droplet size as a percentage of the original droplet's volume in addition to approximately replicating their distribution is important from a simulation point of view, as the parameters for any ejected droplets can be set to follow these values to improve their reliability. The observation of a positive correlation between the volume of a droplet and its speed adds to the realism further. With further agreement found with analytic theories about the evolution of roots of the jet in this study as well, this tells us that both that the BASILISK [19] framework for computational fluid dynamics is reliable and accurate to an intricate level and that the analytic theory is well grounded, despite any modelling simplifications. As such, more complicated situations for which analytic theory may be more difficult to analyse mathematically can have inferences made using this technology. In fact, some discoveries have potentially been revealed through this analysis, notably strong evidence about the evolution of the thickness of the jet through space in the situation of two droplets colliding being $(x^*)^{-7}$ as well as how the distance between the roots of the jet changes through time in the same situation, potentially being at a rate of t^3 . These results may prompt an investigation into the analytic theory of this scenario and, if published, would mean the analysis done in this study would directly contribute towards advancing the field of Wagner theory.

4.4 Methodology Critique and Potential Improvements

The method used to extract a 2-dimensional line from the 3-dimensional data was effective but by no means perfect. The method of taking a 3-dimensional slice and projecting it onto a line meant that information would be lost when stacked together and then compressed, the step where duplicate points were removed was done with a tolerance where the full original dataset could lose a considerable portion of its points. Although enough of the original points still remained after the duplicate removal that the larger features were largely unchanged (and which this project focused on), more intricate details would be lost and if analysis were to be done on a finer simulation and focused on the smaller features, this method may not be so desirable unless incredibly fine slices are taken and the duplicate tolerance is hugely reduced or not done whatsoever. The loss of detail is clear when looking at the finer structures present in the simulations. The ejected droplets and

jet tips in Figure 6, for example, look like unrefined blobs but if a pure slice is taken, say only points with $z = 0$, then their outlines become far clearer and their shapes can be better determined. A possible future analysis could instead use Cartesian lines rather than radial slices. Since the solver works using grids this would make sense, though the nature of splashes being radially symmetric mean that this would be unsuited to the problem. Perhaps even creating a solver which is created in a cylindrical coordinate system could be a useful tool.

The reasoning for removing duplicates was because points incredibly close together caused problems in the algorithms to find the roots, and even then the original curvature definition was changed to be simulation specific despite this since points which got too close would give huge curvature values whilst the surface around it remained fairly smooth. Therefore, a better method of cleaning the data once a slice has been identified would be useful, this could mean adjusting the points such that they are approximately equally spaced but in a way which does not meaningfully change the overall geometry of the desired features. Done properly, this would mean that the curvature definition of the roots of the jet could be used and the code made more generalisable rather than simulation specific.

The projection method also means that for elliptical but non-circular splashes (e.g. a droplet approaching at angle), a sufficiently thin slice must be used that the points captured are approximately radially symmetric. It also assumes that the splash is centred at the origin, for sufficiently simple simulations a transformation of the data can be performed such that this requirement is satisfied; in the case of a droplet approaching a pool at an angle, the frame of reference can be changed such that the droplet is moving directly down and the pool itself is moving instead. However, in a more complex situation such as the edges of two droplets colliding at an angle to each other this may not be so easy and the resulting geometry may not be elliptical at all. Further work could address such possible situations.

When finding the jet roots, the spatial continuity condition ensures that the roots lie on an ellipse in the x - z plane, but there is no assurance that there is such continuity with their y -coordinates. Especially in the case of the colliding droplets, the x and z coordinates of both the top and bottom roots are practically identical so the code may not notice if a root on the other side of the impact was found instead. To combat this, either the root finding should be made more robust or potentially a three-dimensional ellipse could be found instead.

When finding the centre line of the jet, the method only works if the jet is well behaved enough. In cases such as where the jet hits the droplet again this will not work, therefore the jet must be checked before performing to ensure that it works. Whilst not much can be done in the way of fixing such a situation to find a centre line, writing code which can identify that there is an irregularity automatically would be of huge help to a user. Sometimes when identifying the jet, despite the constraints on search direction, the geometry would be such that the search is performed in the wrong direction and maps out the edge of the droplet instead. This can happen if the left root happens to be both the local minimum and closest point to the y -axis. Further robustness measures could have been put in place to prevent this if time and need permitted, but a future project may wish to make code of this nature more user friendly. Moreover, implementation of smoothing algorithms to the centre line may make it more presentable and would make it more tolerant to outlier data points. This would also help the thickness algorithms, sometimes a sharp angle was found and in the second algorithm this could mean that the normal line would pass through the tip of the jet and the nearest points to the line would be found there. Some extra safeguards against such situations would be needed to make the analysis process better automated.

In general, the code was not written in a manner which could be released as a public package. Whilst files were arranged to separate different aspects of the project, the code itself was created for ad hoc use, commonly needing edits when the task at hand changed. Docstrings and comments have been added to the code to help readability, but it was created with the intention of personal use rather than with a user in mind. Creating software in a publishable manner which has more flexible functions, a friendlier user interface and is usable by other splash analysers could be a fruitful endeavour.

5 Conclusion

In this project, analysis has been performed on numerical simulations of high speed water droplet collisions in three dimensions in two canonical situations: a droplet landing on a pool and two droplets colliding directly into each other. The analysis specifically focused on the very early impact stage, within just 20 μ s of impact.

These simulations were performed via the BASILISK open source software. Results gathered have been compared to theoretical derivations, previously performed image analysis and other simulations; with focus on the distribution and behaviour of secondary droplets, and the topological features of the liquid jet which form shortly after impact, including the evolution of the roots of the jet and its thickness over time. The topological details were extracted from a three-dimensional point cloud and a methodology to analyse such features via a projection into two-dimensions was independently devised.

The ejected droplet analysis reaffirmed previous analysis and gave support towards theoretical results, finding that the secondary droplets from these simulations had volumes between around 0.35% and 3.5% of the original droplet volume. It was also found that within each time frame, there was a positive correlation between the volume of the ejected droplets and their speeds. However it was noted that the refinement level was not as high as previously cited simulations, the peak number of identified droplets less than 10% of the number found in the literature. The lower bound for the droplet diameters was also right at the brink of the grid size for the simulation, meaning a further refinement would give better for the lower bound on droplet sizes.

The analysis of the jet has given further evidence towards both previously published theories as well as a developing theory and has even hinted towards a new, previously unobserved behaviour. It was found that in both simulations, the average distance of both roots from the y -axis grew approximately with $O(t^{1/2})$ and the distance between the roots for a droplet on a pool grew with $O(t^{3/2})$. This aligns with previous theoretical results. It was also speculated that the thickness of the jet evolves with $O(r^{-7})$ along its length, where r is the distance from the y -axis. Analysis found that this is likely in the case of two colliding droplets and potentially true in the case of a droplet on a pool, though a more precise formulation of the theory is needed. Finally, it was independently discovered that the distance between the roots of the jet in the case of two droplets colliding appears to grow with $O(t^3)$, prompting new analytic theory to be developed to support this. For detailing the simpler topological details, such as the roots of the jet in the droplet collision simulation, the lower refinement level of $L = 9$ was sufficient, but for more complicated features such as the jet itself and the roots in the droplet on a pool case, a refinement level of $L = 10$ was necessary.

This study has shown the power of numerical simulations in computational fluid dynamics. Further evidence towards previous theories have been provided and evidence towards new theories discovered, and demonstrates the power of the BASILISK framework for computational fluid dynamics. It is clear that the numerical approach will be a part of the forefront of research in this area.

6 Appendices

6.1 Appendix A

The parameters for the simulations are set as follows:

Parameter	Value	Units	Description
ρ_l	999.8	kgm^{-3}	Density of the liquid
ρ_a	1.12	kgm^{-3}	Density of the air
μ_l	8.91×10^{-4}	Pa s	Viscosity of the liquid
μ_a	1.12×10^{-5}	Pa s	Viscosity of the air
σ	7.2×10^{-2}	Nm^{-1}	Surface tension
R_0	5×10^{-4}	m	Droplet radius
u	10	ms^{-1}	Velocity of the droplet
g	9.81	ms^{-2}	Gravity

Table 2: Parameter values for the initial conditions of the simulation.

This is equivalent to a droplet of diameter 1 mm moving at an initial speed of 10 ms^{-1} . At a velocity this high with such a small volume of liquid, this means that both the Reynolds and Weber numbers are incredibly high meaning that analytic theory which approximates these numbers to be infinite can be applied.

6.2 Appendix B

The analysis code assumes data in the form of a Pandas DataFrame [24]. The datasets studied in this project came in the form of a written dat file with columns separated by a blank space. Converting this type of code into a DataFrame is very simple using inbuilt Pandas functions. The columns of the DataFrames should be labelled as `frame`, `time`, `droplet_ID`, `vol_t`, `x`, `y`, `z`, `v_x`, `v_y`, `v_z`.

Two-dimensional data can also be used in this analyser by simply inserting a column of zeros in the z and v_z columns. Similar can be done if the simulation has no velocity information.

6.3 Appendix C

The simulator creates droplets with randomised volumes and in two different zones, these droplets are 'fired' towards each other at a randomised vertical angle. These then travel via simple projectile motion; at each time step it checks whether any droplets are overlapping, if they are then a 'merge event' occurs in which the two droplets combine to become one larger volume and their velocity is updated in accordance with the conservation of momentum. That is

$$\text{Vol}_{\text{new}} = \text{Vol}_1 + \text{Vol}_2,$$
$$v_{\text{new}} = \frac{\text{Vol}_1 v_1 + \text{Vol}_2 v_2}{\text{Vol}_{\text{new}}}.$$

Since Volume $\propto$ Mass, this is equivalent to the usual definition. The code used to create this and an animation to display it are provided in the additional materials [14].

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